LAB 1: REQUIREMENTS ENGINEERING

Food Bank Surplus Redistribution Platform

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- **Requirements Table**

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow donors to post surplus food packages with perishable expiry countdown timers and dietary tags.	High	Pass: Shelter reserves the batch and receives a pickup verification code before expiry. Fail: An expired food batch remains active on the claim board.	Core listing functionality that lets shelters identify and claim food before it spoils.
FR-002	Functional	The system shall allow a verified shelter coordinator to reserve a pickup window for a specific, unclaimed food batch.	High	Pass: When a shelter selects an available time slot, the batch status changes to "Reserved" and is removed from the public claim board within 2 seconds. Fail: The batch remains visible/claimable by other shelters after reservation.	Prevents two shelters from claiming and travelling for the same surplus batch.
FR-003	Functional	The system shall generate a unique pickup verification code upon successful reservation of a food batch.	High	Pass: A unique 6-digit alphanumeric code is generated and shown to both the shelter coordinator and the donor within 5 seconds of reservation confirmation. Fail: No code is generated, or a duplicate code is issued.	Lets the donor confirm the correct shelter representative at handoff, reducing fraud and mix-ups.
FR-004	Functional	The system shall allow shelter coordinators to filter available food batches by dietary tag and by distance from their registered location.	Medium	Pass: Applying a filter (e.g., "vegetarian", "under 5 km") returns only batches matching every selected criterion. Fail: The result list includes any batch that does not match a selected filter.	Helps shelters quickly find batches that fit their clients' dietary needs and pickup capacity.
FR-005	Functional	The system shall automatically remove a food batch from the claim board once its expiry countdown reaches zero and shall notify the posting donor.	Medium	Pass: At expiry, the batch status changes to "Expired", it disappears from the claim board, and the donor receives an expiry notification within 1 minute. Fail: The expired batch remains claimable, or the donor is not notified.	Protects food-safety integrity and prevents shelters from claiming unsafe, expired food.
NFR-001	Nonfunctional (Performance & Security)	The platform shall dispatch instant push notifications to shelters within a 5 km radius when a high-quantity perishable food batch is listed, over an encrypted channel.	High	Pass: Load-test benchmarks confirm notification delivery to all in-radius shelters within 10 seconds of listing at peak simulated load, with all traffic encrypted via TLS 1.2+. Fail: Delivery exceeds 10 seconds for any in-radius shelter, or traffic is sent unencrypted.	Nearby shelters must be alerted fast enough to act before perishable food spoils, without exposing donor/shelter data.
NFR-002	Nonfunctional (Availability & Usability)	The platform shall be available at least 99.5% of the time (excluding scheduled maintenance) and shall let a shelter coordinator complete a reservation in no more than 3 taps after selecting a batch.	Medium	Pass: Uptime monitoring shows ≥ 99.5% availability over any rolling 30-day period, and usability testing shows ≥ 95% of test users complete a reservation within 3 taps. Fail: Uptime falls below 99.5%, or the majority of test users need more than 3 taps.	Because surplus food spoils quickly, the platform must be reliably up and let shelters act with minimal friction.

- **UML DIAGRAM**


Food Bank Surplus Redistribution Platform
Post Surplus Food Batch
Browse / Search Food Batches
Donor Partner
Verify Shelter Credentials
<include>
Shelter Coordinator
Reserve Pickup Window
<include>
Notification Service
<extend>
Generate Pickup Verification Code
Cancel Reservation
Platform Admin
Approve Donor Registration
Remove Expired Batch

- # Use Case

Actors

Primary Actor: Shelter Coordinator
Supporting Actors: Donor Partner, Notification Service

Preconditions

- The Shelter Coordinator is registered and verified on the platform.
- At least one active (non-expired, unreserved) food batch is listed on the claim board.

Postconditions

- Success: The selected batch's status is set to “Reserved” for the chosen pickup window, a unique pickup verification code is generated, and both the Shelter Coordinator and the Donor Partner are notified.
- Failure: The batch remains unreserved and visible on the claim board; no verification code is generated.

Main Success Scenario

1. Shelter Coordinator browses the claim board and selects a listed food batch.
2. System displays batch details: quantity, dietary tags, expiry countdown, and donor pickup location.
3. Shelter Coordinator selects an available pickup window.
4. System verifies the Shelter Coordinator's credentials («include» Verify Shelter Credentials).
5. System checks that the batch has not expired and is not already reserved.
6. System reserves the batch for the selected pickup window.
7. System generates a unique pickup verification code («include» Generate Pickup Verification Code).
8. System sends the confirmation and verification code to the Shelter Coordinator and notifies the Donor Partner.
9. Batch status updates to “Reserved” and is removed from the public claim board.
10. Use case ends successfully.

Alternate Flow – 5a: Batch Already Reserved or Expired

- 5a1. System detects the batch was reserved by another shelter or has expired since selection.
- 5a2. System displays a “Batch no longer available” message to the Shelter Coordinator.
- 5a3. System returns the Shelter Coordinator to an updated claim board.
- 5a4. Use case ends without a reservation being made.